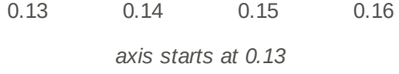
The eight versions of CLR we tested, from raw dependence to the full method

E. coli: strong-evidence regulons, 466 experiments (RegulonDB). Single cells: BEELINE, median over 14 dataset-and-gene-set cases.

Version	Dependence	Context normalization	Softmax	Walk	E. coli AUPR (SC)	BEELINE median ratio
CLR context normalization, scores ranked directly	mutual information	CLR z	—	—	0.136	1.48
CLR-NS CLR on |r| of normal scores (Blom)	|r| of normal scores	CLR z	—	—	*not in the ablation*	**1.60**
|r| (correlation) no normalization, scores ranked directly	|r|	—	—	—	*not in the ablation*	1.20
Attention on |r| softmax + walk, no CLR normalization	|r|	—	softmax	t* = 8	0.143	
Attention on mutual information softmax + walk, no CLR normalization	mutual information	—	softmax	t* = 6	0.143	
Attention on CLR of |r| CLR normalization, then softmax + walk	|r|	CLR z	softmax	t* = 7	0.153	
Attention on CLR-NS CLR-NS scores, then softmax + walk	|r| of normal scores	CLR z	softmax	t* = 5	0.151	no gain over CLR-NS
CLR attention (this work) MI, CLR normalization, softmax + walk	mutual information	CLR z	softmax	t* = 9	**0.158**	

t* = held-out depth, chosen per version; the method comparison itself fixes t = 9.



Compared with (BEELINE median ratio): GENIE3 1.69 PIDC 1.56 GRNBoost2 1.47

scPRINT zero-shot attention: CLR-NS ahead in 7 of 7 datasets (+1.03 log2 AUPR)